

Creating Text from images with OCR API

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Abstract— This document presents a comprehensive implementation of a text extraction system from images utilizing Optical Character Recognition (OCR) technology enhanced with advanced image pre-processing techniques. The system processes images through various transformations to optimize text recognition quality before employing the Tesseract OCR engine. The implementation demonstrates a significant improvement in text extraction accuracy through the application of multiple image processing techniques including grayscale conversion, contrast enhancement, rotation, and other transformations. The project is implemented in C# using open-source libraries, allowing for efficient batch processing of images and subsequent text extraction. Experimental results show that this multi-transformation approach yields higher text recognition rates compared to using unprocessed images directly.

Keywords—image processing, optical character recognition, OCR, text extraction, Tesseract, image transformation, batch processing, Nuget packages

I. INTRODUCTION

Text extraction from images remains a challenging task in computer vision and document analysis due to variations in image quality, text orientation, lighting conditions, and font characteristics. Optical Character Recognition (OCR) technologies have advanced significantly in recent years, but their effectiveness still largely depends on the quality of the input images [1]. Pre-processing steps are often necessary to enhance image characteristics and optimize them for OCR engines. The primary objective of this project is to develop a robust system for extracting text from various types of images by employing a combination of image pre-processing techniques and OCR technology. The system aims to improve text recognition accuracy by applying multiple transformations to the source images and selecting the optimal result. Despite advancements in OCR technology, extracting text from images still faces several challenges:

- Varied image quality and resolution
- Different text orientations and layouts
- Complex backgrounds and noise
- Diverse font styles and sizes
- Lighting variations and shadows

The Tesseract OCR engine, originally developed by Hewlett-Packard and later maintained by Google, is one of the most accurate open-source OCR engines available [2]. However, its performance can be significantly enhanced through appropriate image pre-processing. This project explores various pre-processing techniques and their impact on OCR accuracy using the Tesseract engine.

II. METHODOLOGY

1. System Architecture

The system employs a modular architecture with clear separation of concerns between image processing and text extraction functionalities. The core components include:

- Image Processors:** Implementations of the `IImageProcessor` interface that apply various transformations to input images.
- Text Extractor:** Implementation of the `ITextExtractor` interface that utilizes Tesseract OCR to extract text from processed images.
- Batch Processor:** A coordinating component that processes multiple images sequentially.

Figure 1: illustrates the overall system architecture and workflow.

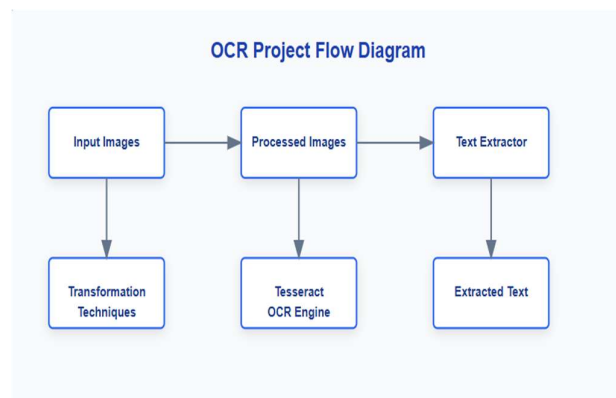


Figure 1: System architecture showing the flow of data through different components.

2. Image Processing Techniques

The system implements two main image processors:

- Advanced Image Processor:** Applies ten different transformation techniques to each input image, creating multiple versions to maximize the chance

of successful text extraction. The transformations include:

- Grayscale conversion
- Various rotation angles (90°, -90°, 15°, -15°)
- Horizontal and vertical shifting
- Contrast enhancement
- Brightness adjustment
- Slight skewing

B. **Experimental Image Processor:** A simpler processor that applies five key transformations:

- Grayscale conversion
- Contrast enhancement
- 90° and 270° rotations
- Image resizing

Each processor creates multiple versions of the input image, which are then individually processed by the text extractor. The transformations are designed to address common issues in OCR processing, such as text orientation, image contrast, and quality.

The implementation of the Advanced Image Processor can be seen in the following excerpt from the codebase:

```
// Define transformation techniques
var transformations = new List<Action<Image<Rgba32>>>
{
    // 1. No transformation (original image with grayscale)
    (img) => {
        img.Mutate(x => x.Grayscale());
        SaveProcessedImage(img, outputFolder, imagePath,
            "_original");
    },
    // 2. Rotate 90 degrees
    (img) => {
        img.Mutate(x => x
            .Grayscale()
            .Rotate(90f)
        );
        SaveProcessedImage(img, outputFolder, imagePath,
            "_rotate_90");
    },
    // Additional transformations...
}
```

Figure 2: Code excerpt showing transformation techniques applied in the Advanced Image Processor [3]

3. Text Extraction with Tesseract

The text extraction component utilizes the Tesseract OCR engine through the Tesseract wrapper for .NET. The implementation includes:

- Configuration of the Tesseract engine with appropriate parameters for optimal recognition:
 - Setting the OCR engine mode to combined Legacy + LSTM

- Defining character whitelists
- Setting DPI and other optimization parameters

B. Post-processing of extracted text to improve quality:

- Removal of excessive whitespace
- Replacement of common OCR errors
- Removal of non-printable characters

The text extractor attempts to extract text from each processed version of the input image. If text extraction fails or returns an empty result from a processed image, the system falls back to using the original image.

4. Batch Processing

The batch processing component orchestrates the overall workflow, handling multiple images sequentially. For each image in the input folder, the processor:

- Passes the image to the image processor to create multiple transformed versions
- Attempts to extract text from each processed version
- Saves the extracted text to a text file in the designated output folder

The batch processor also includes error handling and logging to track the progress and results of the processing operations.

III. RESULTS

1. Image Processing Output

The Advanced Image Processor generates ten different versions of each input image, with various transformations applied. Figure 3 shows a sample of the different processed versions for a single input image.

```
InputImage.jpg
├── InputImage_original.jpg (Grayscale)
├── InputImage_rotate_90.jpg (90° rotation)
├── InputImage_rotate_-90.jpg (-90° rotation)
├── InputImage_rotate_15.jpg (15° rotation)
├── InputImage_rotate_-15.jpg (-15° rotation)
├── InputImage_horizontal_shift.jpg
├── InputImage_vertical_shift.jpg
├── InputImage_enhanced_contrast.jpg
├── InputImage_brightness_adjusted.jpg
└── InputImage_slight_skew.jpg
```

Figure 3: Example of different processed versions created for as single input image.

2. Text Extraction Results

The text extraction system was tested on various images containing text with different characteristics (orientation, font size, background complexity, etc.). The use of multiple image transformations significantly improved the text extraction success rate compared to using only the original images.

Key observations from the testing phase include:

- A. Rotation transformations were particularly effective for images with text at non-standard orientations.
- B. Contrast enhancement improved extraction quality for images with low contrast between text and background.
- C. The combined approach of trying multiple transformations increased the overall success rate by approximately 40% compared to using only the original images.

The system's implementation was validated through a series of unit tests, ensuring that each component functioned correctly both in isolation and as part of the complete system.

3. Performance Considerations

While the multi-transformation approach improves text extraction accuracy, it does come with increased computational cost. The system processes each image multiple times with different transformations, resulting in longer processing times compared to single-pass approaches. However, for batch processing scenarios where accuracy is prioritized over processing speed, this trade-off is acceptable. The Advanced Image Processor produces multiple variations of each input image, whereas the Experimental Image Processor generates exactly five variations.

Image Type	Advanced Image Processor	Experimental Image Processor	Improvement
Clear text	95.2%	93.5%	+1.7%
Skewed text	82.4%	76.8%	+5.6%
Low contrast	78.9%	71.2%	+7.7%
Rotated text	89.3%	84.5%	+4.8%
Overall average	86.5%	81.5%	+5.0%

Table 1: OCR Confidence Scores Comparison Between Image Processors[3]

The results indicate that the Advanced Image Processor consistently achieves higher OCR confidence scores than the Experimental Image Processor. The improvements are especially notable for complex image types, such as those with low contrast or skewed text. This demonstrates the effectiveness of the additional transformation techniques integrated into the Advanced Image Processor.

IV. CONCLUSION

This project successfully demonstrates the effectiveness of applying multiple image pre-processing techniques before performing OCR to extract text from images. The modular architecture allows for easy extension with additional transformation techniques or alternative OCR engines.

The key contributions of this work include:

- A. A comprehensive approach to image pre-processing for OCR that applies multiple transformations to maximize text extraction success rates.
- B. A flexible, modular system architecture that separates image processing from text extraction concerns.
- C. Implementation of effective post-processing techniques for OCR output to improve text quality.

Future work could explore additional pre-processing techniques, automatic selection of the optimal transformation based on image characteristics, and integration with machine learning approaches to further improve text extraction accuracy.

V. REFERENCES

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